

# Temporal-Support Identifiability in Event-Detection Evaluation Pipelines

Anonymous CVPR submission

Paper ID \*\*\*\*\*

## Abstract

001 *An event-detection benchmark identifies each scored sam-*  
002 *ple by one nominal timestamp, but a recurrent detector*  
003 *forms its prediction from a longer history, and the bench-*  
004 *mark does not record how long. We state the missing re-*  
005 *quirement as a property of the protocol, temporal-support*  
006 *identifiability: the benchmark variables must uniquely de-*  
007 *termine the temporal evidence supporting a reported pre-*  
008 *diction. This makes effective temporal support part of the*  
009 *benchmark specification rather than a hidden property of*  
010 *one evaluation implementation. A released SSM-ViT eval-*  
011 *uation pipeline provides a direct counterexample. At the*  
012 *same evaluated timestamp, with frames, weights, labels and*  
013 *evaluator fixed, moving only the chunk boundary changes*  
014 *the recurrent history from one to twenty-one windows within*  
015 *the pooled score, and changes S5-B by 4.20 mAP over*  
016 *all labeled boxes and by  $5.23 \pm 0.49$  mAP on the velocity-*  
017 *evaluable subset. Because identifiability is a uniqueness*  
018 *property, one controlled pair in which identical recorded*  
019 *protocol variables yield different effective support and dif-*  
020 *ferent scores is sufficient to show that the protocol under-*  
021 *specifies the score. Two supporting diagnostics bound the*  
022 *interpretation without being used as recurrent-support es-*  
023 *timators: on the predictor side, the newest RVT input win-*  
024 *down has an ablation-sensitivity centroid 23.81 ms before the*  
025 *label instant and a nominal-time-matched regression coeffi-*  
026 *cient of  $-2.40 \pm 7.18$  ms; on the label side, within-exposure*  
027 *event timing contains an illumination-locked component.*  
028 *Over all labeled boxes the validation-split ordering of the*  
029 *5 released checkpoints is itself reproduced in only 42.5 %*  
030 *of sequence-bootstrap replicates, so that ordering cannot*  
031 *certify a timing convention as immaterial. Timing compar-*  
032 *isons between recurrent detectors therefore require the sup-*  
033 *port that produced each score to be stated and equalized,*  
034 *a condition an event-detection protocol can record with its*  
035 *labels.*

## 1. Introduction

Two recurrent detectors evaluated at the same nominal timestamp are not comparable in time unless the history each one actually used is also identified.

An event-detection benchmark defines a sample by a nominal timestamp. A prediction  $\hat{s}$  is scored against the ground truth  $s^*$  carried at query time  $t$  through a function  $\rho(\hat{s}, s^*)$  of position, shape and class. Beyond selecting the label at  $t$ , that function contains no variable for the temporal evidence behind either argument, so any temporal discrepancy enters the score disguised as displacement.

A recurrent detector does not read only the instant the benchmark names. It computes  $\hat{s}$  from an input window and from state carried across earlier windows, and the label it is compared against comes from an exposure of finite width, an annotator or tracker working on the exposed image, and a geometric transfer into another sensor's frame. The evidence on both sides is an interval, while the protocol records an instant.

Three terms carry this throughout. The *nominal timestamp* is the instant at which the protocol pairs a prediction with a label. The *effective temporal support* is the past that can influence the prediction, including state carried by recurrence. The *label-acquisition support* is the interval over which the label's evidence was acquired. A nominal timestamp determines neither of the other two uniquely, and a temporal descriptor may be reconstructable from released artifacts yet remain unreported or heterogeneous within one pooled score. We call this protocol property temporal-support identifiability: whether the released protocol and artifacts uniquely determine the temporal quantities a reported score represents.

We test it with a paired intervention on the released SSM-ViT evaluation pipeline. In that implementation the nominally carried state is inert across evaluation chunks, so the recurrent history available to a prediction is a deterministic function of chunk position and ranges from one to twenty-one windows inside one pooled score. Moving every chunk boundary 5 windows later re-scores the same

frames under a longer history while frames, weights, ground truth, evaluator and the evaluated timestamp are held fixed (3 574 paired boxes, cluster bootstrap over 406 sequences).

Effective temporal support alone changes S5-B by 4.20 mAP over all labeled boxes and  $5.23 \pm 0.49$  mAP on the velocity-evaluable subset (Sec. 3); a second checkpoint of the same released pipeline moves by 4.10 and 4.81 points. Since temporal-support identifiability is a uniqueness property, this controlled violation is sufficient to show that the nominal timestamp does not uniquely specify the temporal evidence represented by that score.

## Contributions.

- *Temporal-support identifiability*: an evaluation property separating the nominal timestamp a protocol records from the effective temporal support and label-acquisition support a reported score represents.
- A paired intervention that isolates recurrent support: the frames, weights, labels, evaluator, and evaluated timestamp are fixed, while only the recurrent history changes.
- Its measured consequence: support the protocol does not record moves a pooled leaderboard-style score by 4.20 mAP overall and 5.23 mAP on the velocity-evaluable subset, while the same benchmark reproduces its validation-split ordering over all labeled boxes in only 42.5 % of sequence-bootstrap replicates; predictor-side and label-side audits delimit the result.

Work on event-detection timing takes that index as given and improves what is measured against it: sensor latency is calibrated away [25, 29], compute latency is folded into average precision [9, 30, 35], and the temporal window is redesigned [24, 26]. Each changes what the predictor does at a fixed benchmark index, whereas the object measured here is that index itself and the support a score reported under it pools. The two datasets below instantiate this protocol condition; how often it holds elsewhere is not measured.

**Supporting audit: predictor-side support.** The event branch examined here [24] consumes a window of 50 ms in ten bins ending on the label instant (Sec. 4.2), so under uniform weighting its evidence is centered 25 ms earlier; ablating each bin on real validation samples puts the ablation-sensitivity centroid at  $-23.81$  ms (Sec. 4.3), the newest window carrying 22.7% of the summed window-ablation sensitivity over a 1000 ms horizon and preceding windows 77.3%. The regression-based temporal coefficient, estimated on nominal-time-matched detections under Eq. 2 with a placebo term and a sign-flip control, is  $-2.40 \pm 7.18$  ms, sits 26.2 ms from that centroid and is compatible with zero: it is an association between nominal-time-matched detections, not an estimate of recurrent support, and the nominal timestamp encodes neither (Secs. 4.2–4.5).

**Supporting audit: label-acquisition support.** Across

the eighteen DSEC training sequences whose exposure files resolve, published exposure width varies by a factor of 127 against a frame period fixed at 50 ms, with 32.5 % of frames pinned at an auto-exposure ceiling in six sequences (Fig. 3). Those sequences carry the harmonic series of a 100 Hz mains flicker, and between 95.4 % and 100.0 % of labeled boxes exceed the ninety-ninth percentile of the off-frequency local excess measured on the same events, so within-exposure event timing is not attributable to object timing without separating that component (Secs. 5–5.4).

Identifiability under the metric is then read in the metric’s own units. On the velocity-evaluable subset, displacing ground-truth centers across the measured label-to-centroid interval changes mAP by 0.04 points without reordering the 5 released checkpoints, an effect at the benchmark’s sampling-resolution floor. Table 2 relates this resolution to the benchmark’s speed composition (Supplement Secs. 11–12).

## 2. Related work

**Sensor and runtime latency.** Prior work calibrates sensor latency [25, 29], incorporates runtime latency into average precision [9, 30, 35], or proposes temporal tolerance [36]. These methods address the time from an event to an output and take the benchmark’s temporal index as given. We instead measure that index, testing whether a released benchmark specification uniquely determines the temporal quantities behind its score.

**Temporal windows and benchmark identifiability.** RVT and SSM-ViT redesign or parameterize temporal windows [24, 26], while adaptive-window methods compare alternative partitions [28]. Event–RGB frequency mismatch and exposure duration have also been noted [22, 23, 37]. The closest accepted work above calibrates sensor or runtime latency or redesigns temporal windows, each holding the benchmark’s temporal index fixed. We make that index the measured quantity, reading the released per-frame windows and exposures and separating predictor support, output-time interpretation, and label-acquisition support.

**Benchmark reliability and evaluation analysis.** Prior work studies detector calibration, stratified model evaluation, label-induced benchmark instability, and event-vision evaluation protocols [31–34]. These analyses examine uncertainty or error in reported performance, whereas our focus is the temporal support and acquisition semantics that a nominal detection timestamp leaves implicit.

**Label time and attribution.** Waymo and annotation studies estimate per-object capture or annotation offsets [5–7]. Event datasets expose per-box time fields, but their de-

|     |                                                                     |     |
|-----|---------------------------------------------------------------------|-----|
| 175 | degrees of freedom are not necessarily specified. LET-3D-AP         | 223 |
| 176 | and TIDE/DETAD/HOTA decompose or bucket spatial error               | 224 |
| 177 | while taking the label clock as given [1, 11–14].                   | 225 |
| 178 | <b>Event benchmarks and exposure.</b> Gen1, 1 Mpx, DSEC-            | 226 |
| 179 | Det and Ev-3DOD use asynchronous measurements, rec-                 | 227 |
| 180 | tified frames, or interpolated labels [16, 19–21]. LEOD             | 228 |
| 181 | establishes causal input windows [27]. The present work             | 229 |
| 182 | reads the released window and exposure artifacts as proto-          | 230 |
| 183 | col evidence and tests whether the resulting temporal quan-         |     |
| 184 | tities are uniquely specified.                                      |     |
| 185 | <b>3. Primary evidence: the same-frame chunk-</b>                   |     |
| 186 | <b>boundary intervention</b>                                        |     |
| 187 | The state the SSM-ViT release carries between evaluation            |     |
| 188 | chunks does not affect any of its outputs: it enters as a multi-    |     |
| 189 | plicative factor on the first coefficient of an associative scan    |     |
| 190 | on which the outputs do not depend. Under the released              |     |
| 191 | streaming evaluation this bounds the recurrent history avail-       |     |
| 192 | able to a detection to the current 21-window chunk, so that         |     |
| 193 | history ranges from one window to 21 with the detection’s           |     |
| 194 | position in the chunk, and the reported number averages             |     |
| 195 | over all 21 positions. Chunk position is a deterministic            |     |
| 196 | function of the release’s own index files and is recorded           |     |
| 197 | nowhere in the evaluation protocol.                                 |     |
| 198 | <b>The same-frame intervention.</b> The intervention moves          |     |
| 199 | every chunk boundary 5 windows later and changes nothing            |     |
| 200 | else: the same frames, the same weights, the same labels,           |     |
| 201 | the same evaluator, and the evaluated timestamp of each             |     |
| 202 | frame unchanged. 3 574 labeled boxes are scored twice               |     |
| 203 | under this shift, once with one to four windows of recur-           |     |
| 204 | rent history and once with seventeen to twenty, and are             |     |
| 205 | compared paired within frame with a cluster bootstrap over          |     |
| 206 | the 406 sequences. The long-history arm gains $5.23 \pm 0.49$       |     |
| 207 | mAP points for S5-B and $4.81 \pm 0.52$ for S5-S on the             |     |
| 208 | velocity-evaluable subset, and over every labeled box the           |     |
| 209 | S5-B gain is $4.20 \pm 0.36$ points. Mean detection confidence,     |     |
| 210 | which uses no labels, moves with it by $0.0579 \pm 0.0048$ and      |     |
| 211 | $0.0509 \pm 0.0035$ . Fig. 1 (bottom) draws one such pair. No       |     |
| 212 | quantity the protocol records distinguishes the two arms.           |     |
| 213 | <b>The observational contrast across chunk positions.</b> Re-       |     |
| 214 | constructing chunk position over 20 296 labeled frames of           |     |
| 215 | the released evaluation and scoring every labeled box, S5-B         |     |
| 216 | reaches 34.445 mAP on the frames given one to four win-             |     |
| 217 | dows and 40.226 on those given seventeen to 21 (Fig. 1,             |     |
| 218 | top). This contrast compares different frames, so it is dif-        |     |
| 219 | ferenced against the three RVT checkpoints, evaluated on            |     |
| 220 | the same frames with a state that crosses chunk bound-              |     |
| 221 | aries; that controls for frame-position effects shared with         |     |
| 222 | them and leaves $5.32 \pm 0.53$ points for S5-B ( $z = 10.0$ )      |     |
|     | and $4.37 \pm 0.61$ for S5-S ( $z = 7.2$ ), against placebo differ- |     |
|     | ences among the RVT checkpoints of at most 0.67 points              |     |
|     | ( $z = 1.56$ ). Permuting chunk position within each sequence       |     |
|     | puts both SSM values at least 7.6 standard deviations out-          |     |
|     | side their own null and every placebo inside it. The two            |     |
|     | readings agree in sign and magnitude, and the paired inter-         |     |
|     | vention carries the claim because it holds the frames fixed         |     |
|     | (Supplement Sec. 13).                                               |     |
|     | <b>The control, measured rather than assumed.</b> That dif-         |     |
|     | ferencing treats the RVT checkpoints as a control because           |     |
|     | their state survives a chunk boundary. The same-frame in-           |     |
|     | tervention applied to RVT measures that property. Driven            |     |
|     | on the SSM chunk grid, RVT reproduces its stored per-               |     |
|     | window detections bit for bit at zero shift, and the same           |     |
|     | 3 574 frames are re-scored with every boundary moved 5              |     |
|     | windows later. With the released state carried across the           |     |
|     | boundary the paired change is $-0.01 \pm 0.03$ points for RVT-      |     |
|     | s ( $z = -0.3$ ) and $0.05 \pm 0.04$ for RVT-b ( $z = 1.3$ ). Drop- |     |
|     | ping the state at every chunk start, the condition the SSM          |     |
|     | release is already in, moves the same weights on the same           |     |
|     | frames by $5.39 \pm 0.45$ ( $z = 12.0$ ) and $6.91 \pm 0.64$ points |     |
|     | ( $z = 10.9$ ), with random-frame nulls centered within 0.112       |     |
|     | points of zero. The boundary effect follows whether the re-         |     |
|     | current state survives the boundary, not the architecture, and      |     |
|     | the control the difference-in-differences relies on holds as a      |     |
|     | measurement (Supplement Sec. 13).                                   |     |
|     | <b>What the pooled score reports.</b> One released number           |     |
|     | therefore pools detections whose recurrent history differs by       |     |
|     | a factor of 21. Restricting evaluation to the frames receiving      |     |
|     | 17–21 windows yields an S5-B score 1.75 points above the            |     |
|     | pooled released-evaluation score. Two systems compared              |     |
|     | under this protocol at the same nominal timestamps need             |     |
|     | not have been given the same temporal evidence, and the             |     |
|     | protocol records no variable by which a reader could tell.          |     |
|     | <b>4. Supporting audit: predictor-side temporal</b>                 |     |
|     | <b>descriptors</b>                                                  |     |
|     | <b>4.1. Identifiability and the timing estimands</b>                |     |
|     | A benchmark releases one tuple per evaluated instant: a             |     |
|     | nominal timestamp, the labels asserted at it, the predictions       |     |
|     | scored against them, and the score. Three temporal quan-            |     |
|     | tities are latent in that tuple and are not recorded in it: the     |     |
|     | temporal support of the predictor’s input and carried state,        |     |
|     | the effective state time its output refers to, and the acqui-       |     |
|     | sition support of the labels. Call such a quantity <i>protocol-</i> |     |
|     | <i>identifiable</i> when the released protocol and artifacts deter- |     |
|     | mine it uniquely. Under this definition, input and exposure         |     |
|     | intervals are protocol-identifiable, whereas effective predic-      |     |
|     | tor timing, recurrent context represented by a pooled score,        |     |
|     | and within-exposure attribution are not.                            |     |

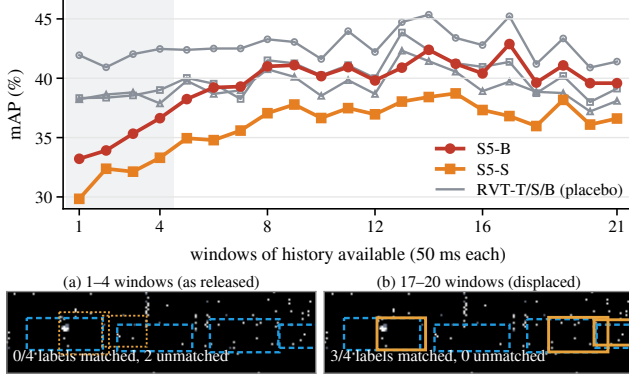


Figure 1. **Top:** benchmark score against the recurrent history the released streaming evaluation made available, at each of the 21 chunk positions, over every labeled Gen1 box with no displacement applied. Shaded: the one-to-four-window block the release scores together with the rest. Between that block and frames given seventeen or more windows, the two SSM checkpoints rise by 5.78 and 4.83 points; the three RVT checkpoints, scored on the same frames with state that crosses chunk boundaries, rise by at most 0.80. **Bottom:** one frame of the same-frame intervention on S5-B. Changed: the chunk boundary, and with it the recurrent history behind the detection. Fixed: frame, weights, labels (dashed), evaluator and evaluated timestamp. Solid: a detection matched to a label of its class at  $\text{IoU} \geq 0.5$ ; dotted: unmatched; drawn above confidence 0.3 and scored at every confidence. The example is illustrative; over all 3574 frames the same intervention changes S5-B by 4.20 mAP over all labeled boxes and  $5.23 \pm 0.49$  mAP on the velocity-evaluable subset (Supplement Sec. 13).

Let a predictor  $P$  emit a state at query time  $t$ . A signed effective-weight model uses bin centers  $c_k$  and weights  $a_k$  with  $\sum_k a_k = 1$ , giving the effective timestamp offset  $\tau_P = \sum_k a_k c_k$  under constant velocity. The empirical ablation profile instead assigns non-negative sensitivities  $s_k \geq 0$  to bins and has centroid  $c_s = \sum_k s_k c_k / \sum_k s_k$ . These are distinct quantities: zero-ablation sensitivity depends on both the predictor and the input, and is not in general a decomposition of temporal weights. Sections 4.2–4.3 measure this operational input-sensitivity profile over the newest window. Section 4.4 estimates the regression coefficient  $\tau$  under Eq. 2. Interpreting  $-\tau$  as an effective state-time offset requires the absence of residual velocity-proportional spatial bias. Section 4.5 compares  $-c_s$  with the regression coefficient  $\tau$ , equivalently  $c_s$  with the regression-implied timestamp  $\tau_P = -\tau$ .

## 4.2. The input window and its indexing instant

Two released files fix the input side. The dataset configuration `config/dataset/gen1.yaml` of the detector’s own repository [24] names the representation `stacked_histogram-dt=50_nbins=10`, a window of 50 ms divided into ten bins of 5 ms. Line 405 of `scripts/genx/preprocess_dataset.py` constructs the array `ev_repr_timestamps_us_end` by counting backwards from a label timestamp in steps of the window duration, and extends it between consecutive la-

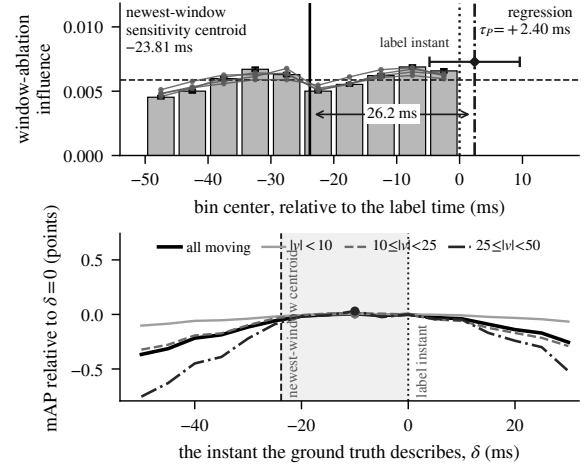


Figure 2. Input sensitivity and the regression-based temporal coefficient summarize the same nominal timestamp without being equivalent, and are not competing estimates of a common latent timestamp; their point estimates are 26.2 ms apart. Panel (b) shows that average precision on the velocity-evaluable subset is nearly invariant over the measured 23.81 ms label-to-centroid displacement. The 26.2 ms separation is not an estimate of architectural latency. (a) Window-ablation influence of each of the ten bins of the released `rvt-t` input window, over 480 Gen1 validation samples with the recurrent state warmed eight steps, with sample-level standard errors. Error bars are descriptive standard errors across the sampled instances and are not used for inference. Uncertainty of the ablation-sensitivity centroid is evaluated by sequence bootstrap in Supplement Sec. 2. Dashed: the mean over bins. Dotted: the label instant. Solid: the ablation-sensitivity centroid. Dash-dot, with its standard error shaded: the regression-implied timestamp under Eq. 2,  $\tau_P = -\tau$ , of Sec. 4.4. Gray: the four other checkpoints, each rescaled to its own mean. (b) Average precision against the temporal center displacement  $\delta$  applied to the ground-truth boxes, by object speed in px/s, each curve relative to its own value at the label instant, with its maximum marked. The shaded band runs from the centroid to the label instant, spanning 0.61 px at the ninetyeth percentile of label speed against an annotation dispersion of 0.6353 px (Supplement Sec. 3). The mAP span is 0.37 points over all moving boxes and 0.78 points in the 25–50 px/s stratum, where the centroid displacement costs 0.08 points.

bel timestamps. Each representation is therefore indexed by the end of its window, and those ends are placed on label times. A window attached to a label at time  $t$  holds the events of  $[t - 50 \text{ ms}, t]$ , whose uniform weighting centroid falls 25 ms before the instant  $t$  at which the label asserts a state. We read both statements from the released source, and the 25 ms figure is a reference point instead of a property of the trained network.

## 4.3. The measured influence profile

The newest-window ablation-sensitivity centroid is computed from the measured bin-ablation sensitivities of a released checkpoint. We take 480 samples of the released Gen1 validation split with the released `rvt-t` checkpoint, warm the recurrent state for eight steps, and ablate each bin in turn. For sample  $i$ , let  $Y_i$  denote the emitted detection ten-

313 sor from the unablated input and  $Y_i^{(-k)}$  the tensor obtained  
314 after zeroing temporal bin  $k$  while keeping the incoming re-  
315 current state identical. We define the ablation sensitivity, or  
316 influence, of bin  $k$  as

$$s_k = \frac{1}{N} \sum_{i=1}^N \frac{\|Y_i^{(-k)} - Y_i\|_2}{\|Y_i\|_2}, \quad N = 480, \quad (1)$$

$$c_s = \frac{\sum_k s_k c_k}{\sum_k s_k},$$

318 where  $c_k$  is the center time of bin  $k$  relative to the label in-  
319 stant. The ablated forward pass receives the same incoming  
320 recurrent state as the corresponding reference pass, and its  
321 returned state is discarded. The profile is a property of the  
322 current window and not of a perturbed history.

323 The profile is balanced between the two halves of the  
324 window (Fig. 2a) and yields an ablation-sensitivity cen-  
325 troid of  $-23.81$  ms, compared with  $-25.00$  ms for uni-  
326 form weighting. The released configuration therefore places  
327 the measured centroid near the window reference with-  
328 out implying uniform network sensitivity. Across zero,  
329 mean, cross-sample replacement, and gradient instruments,  
330 the centroid spans  $-26.35$  to  $-22.35$  ms (Supplement  
331 Sec. 2). The reported centroid applies to the determinis-  
332 tic 12-sequence subset described in the supplement. This  
333 is the current window’s window-ablation sensitivity profile  
334 with the recurrent state held fixed, not a decomposition of  
335 the full temporal support.

336 **Recurrent support beyond the newest window.** Over a  
337 1000 ms horizon, the newest window accounts for 22.7% of  
338 the summed window-ablation sensitivity, while the remain-  
339 ing 77.3% lies in preceding windows. The horizon centroid  
340 shifts from  $-161$  ms at 500 ms to  $-299$  ms at 1000 ms, so  
341 the measured range does not determine a converged full-  
342 history centroid. Supplement Sec. 14 characterizes the tail  
343 and its fill controls.

344 **Across the released checkpoints and a second architec-**  
345 **ture.** RVT releases three Gen1 checkpoints differing only  
346 in embedding width, attention head dimension and FPN  
347 depth. SSM-ViT [26] releases two more in which each  
348 stage’s ConvLSTM is replaced by an S5 state space layer, at  
349 9.68 and 18.19 M parameters against RVT’s 9.87 and 18.54.  
350 It distributes RVT’s own preprocessed Gen1 and inherits its  
351 window construction. On the same sequences, and with  
352 each driven as its own release specifies, all 5 checkpoints  
353 put the newest window’s ablation-sensitivity centroid be-  
354 tween  $-23.76$  and  $-24.72$  ms, none further than 1.24 ms,  
355 or 25 % of one bin, from the uniform-weight value.

356 The support within the newest window is not fixed across  
357 capacities: over 576 samples of the three RVT capacities,

the newest window’s share of a 1000 ms horizon falls from  
22.7 % to 19.0 % and the sensitivity centroid moves from  
 $-299$  to  $-339$  ms.

#### 4.4. Regression-based temporal coefficient

The influence profile is a statement about the network’s in-  
put. Even a predictor with uniform local sensitivity can emit  
a state describing a later instant, and that quantity is mea-  
sured from the same checkpoint’s own detections.

A detector reporting the state an object occupied  $\tau$  ear-  
lier incurs the physical displacement  $-\tau v$ , where  $v$  is the  
latent true velocity. We observe only the finite-differenced  
label velocity  $\tilde{v} = v + \eta$  and therefore fit Eq. 2 using  $\tilde{v}$ . Thus  
 $\tau = -\tau_P$  on the axis of Sec. 4. With  $\hat{c}$  the predicted cen-  
ter,  $c^*$  the label center,  $A$  the area of the ground-truth box,  
 $\hat{u} = \tilde{v}/\|\tilde{v}\|$  and  $R$  a quarter turn,  $b$  is a two-dimensional  
intercept, fitted globally or as per-sequence intercepts in the  
reported specification,

$$\hat{c} - c^* = b - \tau \tilde{v} + \rho R \tilde{v} + \gamma \sqrt{A} \hat{u} + \varepsilon \quad (2)$$

is fitted by stacking both components of every match into  
one least squares problem, with standard errors clustered  
by sequence. It is the temporal calibration of [8] carried to  
a box center, taken as a slope, not as the per-object ratio  
 $-\langle \hat{c} - c^*, \hat{u} \rangle / \|\tilde{v}\|$ , whose variance diverges as its denomi-  
nator approaches zero. For comparison, the per-object ra-  
tio evaluated on the same rows has a mean of  $+24.22 \pm$   
 $27.73$  ms, on the hypothesis evaluated in Sec. 4.5, with an  
interquartile range of 430 ms. Written this way, Eq. 2 fits  
the two-dimensional residual directly with  $\tilde{v}$  as a regressor.  
Unlike the per-object ratio, it does not first project the resid-  
ual onto  $\hat{u}$  and divide by  $\|\tilde{v}\|$ . The reported full specifica-  
tion adds four coordinate/size regressors and per-sequence  
intercepts to the terms displayed in Eq. 2. The coefficient  
is a temporal offset only if no velocity-proportional spatial  
bias remains after these controls. Such a bias is indistin-  
guishable from  $\tau$  in Eq. 2.

$R \tilde{v}$  is the label velocity rotated a quarter turn. Because  
a pure temporal offset acts along  $\tilde{v}$ , not  $R \tilde{v}$ ,  $\rho$  serves as a  
negative-control coefficient. A nonzero value implies resid-  
ual spatial misspecification.  $\sqrt{A} \hat{u}$  absorbs a displacement  
along travel proportional to box side, which is needed be-  
cause that residual grows with box size and box size corre-  
lates with label speed at  $+0.139$  on this split.

**The finite difference used for the velocity.**  $\tilde{v}$  is finite  
differenced from labels. Label noise enters forward and  
backward velocity differences with opposite shared-noise  
signs. The centered difference excludes the current label  
center and removes this direct same-frame term. Supple-  
ment Secs. 1 and 4 derive and calibrate this effect, so the  
reported regression uses the centered estimator.

| Specification                                         | $\tau$ (ms)       | placebo $\rho$ (ms) |
|-------------------------------------------------------|-------------------|---------------------|
| <i>velocity estimator, no geometric controls</i>      |                   |                     |
| forward difference                                    | $-14.68 \pm 4.99$ | $-6.53 \pm 1.68$    |
| backward difference                                   | $+45.65 \pm 5.59$ | $-6.15 \pm 1.71$    |
| centered difference                                   | $+17.51 \pm 5.06$ | $-7.76 \pm 1.92$    |
| <i>controls added, centered difference throughout</i> |                   |                     |
| + box side along travel                               | $+14.62 \pm 5.47$ | $-7.75 \pm 1.91$    |
| + four geometric terms                                | $-7.62 \pm 7.01$  | $-4.12 \pm 1.90$    |
| + per-sequence bias                                   | $-2.40 \pm 7.18$  | $-2.66 \pm 1.71$    |

Table 1. The regression-based temporal coefficient  $\tau$  of the released `rvt-t` checkpoint over 22 534 matched boxes of 376 Gen1 validation sequences, and the placebo  $\rho$  beside it, the coefficient on the rotated label velocity, which no temporal offset can reach. Standard errors are clustered by sequence. The forward and backward fits above are calibrated in Supplement Sec. 4.

**Regression dataset and matching.** Rows are detections matched to nominal-time ground truth at  $\text{IoU} \geq 0.3$  with centered label velocities available, leaving 22 534 matches from 376 sequences. The estimate is conditional on this matched subset. Supplement Sec. 5 reports association-rule sweeps, including overlap-free matching.

Table 1 is a specification ladder, and the specification was chosen by the placebo, not by  $\tau$ . Under Eq. 2 with only the box-side term,  $\tau = +14.62 \pm 5.47$  ms, but  $\rho$  is  $-7.75$  ms at 4.0 standard errors, which indicates a term omitted from Eq. 2 that  $R\tilde{v}$  is a proxy for. A split by the sign of the horizontal velocity locates it:  $\rho$  changes sign across that split, at  $-27.99$  ms and  $+21.84$  ms, which a rotational term would not do. Most label motion here is horizontal, so  $R\tilde{v}$  maps a horizontal speed onto a vertical residual. The rotated regressor represents a bias that differs between oncoming and same-direction traffic. Entering box height and image row along  $\hat{y}$  and box width and image column along  $\hat{x}$ , and then a constant vector per sequence, returns the placebo to  $-2.66$  ms at 1.6 standard errors. That is the specification reported, and the criterion that selected it is a coefficient on a regressor orthogonal to  $\tilde{v}$ .

Under the reported specification,  $\tau = -2.40 \pm 7.18$  ms, with a sequence-bootstrap 95% interval of  $[-17.0, +11.5]$  ms. Across geometric-control specifications,  $\tau$  spans  $-20.2$  to  $+11.1$  ms. Because the reported specification was selected using the orthogonal placebo coefficient, its nominal standard error does not include specification-selection uncertainty. We treat the point estimate descriptively and use the full specification sweep to assess robustness.

#### 4.5. Relating input-sensitivity timing to the regression-based temporal coefficient

The comparison combines measurements under different recurrent-state conditions. The centroid uses an eight-step warm-up with the incoming state fixed during ablation. The regression uses detections from the validation stream

with the released state carried across frames. The 26.2 ms numerical separation is descriptive only: the two quantities are distinct operational summaries and are not competing estimates of a common latent timestamp. Under the reported specification and its annotation-error assumptions, the point-estimate difference is 26.2 ms. The regression coefficient has a standard error of 7.2 ms. This is a regression-coefficient result instead of a latency estimate of this architecture. Expressed on Eq. 2’s sign convention, the newest-window centroid corresponds numerically to  $\tau = +23.81$  ms. This comparison relates two operational estimands and does not assume that Eq. 2 must recover the ablation centroid. Thus, 23.81 ms is the measured interval between the label instant and the newest-window ablation-sensitivity centroid, whereas 26.2 ms is the point-estimate difference between that centroid and the reported regression coefficient. The measured horizons do not determine a converged full-history centroid. The variation across specifications is substantial relative to the reported standard error and is not captured by that standard error.

The detector is trained against the ground truth at the label time on a window ending there, so a measured offset near zero is consistent with the training objective indexed at the label time. The objective does not enforce the interval, and the instant the output describes and the centroid of the evidence used to produce it are different quantities that one benchmark timestamp stands for. At the ninetieth percentile of label speed on this split, 25.7 px/s, the 23.81 ms is 0.61 px of displacement, comparable to the annotation dispersion of the labels themselves (0.6353 px), and at the ninety-ninth it is 1.29 px. That ablation-sensitivity centroid and the regression-based temporal coefficient are measured from released artifacts alone.

**Effect on benchmark mAP.** This experiment is not a test of whether temporal support matters in general. It asks whether Gen1’s particular speed composition makes the measured RVT interval visible to its current mAP. We translate each ground-truth box center by  $\delta\tilde{v}$ , keeping its size and class fixed, a first-order surrogate for temporal displacement under local constant velocity. The released predictions remain unchanged, since a detection has no velocity of its own. Over 20 296 frames, 68.7 % of the 40 698 ground-truth boxes carry a centered velocity. Ground-truth boxes without a centered velocity are marked ignored during evaluation, and detections matched to ignored boxes are excluded from TP/FP accounting. The resulting mAP against  $\delta$  is smooth with its maximum at  $-10$  ms (Fig. 2b). Along the axis of Eq. 2 that is  $\tau = +10$  ms, and mAP varies by at most 0.04 points over the interval. This descriptive curve does not by itself identify a preferred temporal value.

The full  $-60$  to  $+60$  ms sweep moves mAP by 0.9 points. On the  $-50$  to  $+30$  ms grid summarized in Table 2,

| speed (px/s) | share (%) | med. speed | centroid (px) | $\Delta$ mAP | span |
|--------------|-----------|------------|---------------|--------------|------|
| all moving   | 100.0     | 4.0        | 0.10          | 0.04         | 0.37 |
| < 10         | 72.1      | 2.1        | 0.05          | 0.02         | 0.11 |
| 10–25        | 19.2      | 15.0       | 0.36          | 0.01         | 0.33 |
| 25–50        | 7.5       | 32.3       | 0.77          | 0.08         | 0.78 |
| $\geq 50$    | 1.2       | 57.4       | 1.37          | 0.00         | 0.37 |

Table 2. The sweep of Fig. 2b by object speed over the 27 943 ground-truth boxes with centered velocity. Other boxes are ignored.  $\Delta$ mAP is the cost of shifting the ground truth to the newest-window ablation-sensitivity centroid at  $-23.81$  ms. Span is the mAP range over the  $-50$  to  $+30$  ms sweep. Both values are reported in points.

the all-moving span is 0.37 points. Displacing the ground truth to the window’s centroid costs 0.0004 absolute mAP (0.04 percentage points, 0.12 % relative to 0.3346). Average precision on the velocity-evaluable subset is nearly invariant over the measured label-to-centroid displacement, and at the common  $-23.81$  ms displacement the ordering of all 5 released checkpoints is unchanged on the validation split. Scoring each checkpoint at its own preferred displacement likewise leaves the validation-split ordering unchanged in the reported strata. Both statements are bounded by the stability of the ordering under sequence resampling at this scale. The smallest timing difference that would reorder an adjacent pair is 43.7 ms over all moving boxes,  $1.83\times$  the measured interval, and 11.2 ms in the 25–50 px/s stratum. Separately, on every labeled box instead of the velocity-evaluable subset the displacement acts on, the benchmark’s own cluster resampling holds the validation-split ordering in only 42.5 % of replicates, its tightest adjacent pair separated by 0.050 points against a paired standard error of 0.53. A common displacement sweep reveals one localized inversion between `rvt-b` and `S5-S` in the 25–50 px/s stratum. Table 2 locates the flatness in that composition: 72.1 % of the moving boxes travel below 10 px/s, where the displacement is 0.05 px, and the sweep deepens to 0.78 points between 25 and 50 px/s, where it is 0.77 px, past the labels’ own annotation dispersion of 0.6353 px (Supplement Sec. 3). The near invariance therefore characterizes Gen1’s metric resolution, not the absence of a temporal discrepancy.

## 5. Supporting audit: label-acquisition timing within one exposure

### 5.1. Exposure and labels

The predictor-side analysis concerns the temporal support of released detector inputs and outputs. DSEC provides a separate exposure-side analysis: published exposure intervals bound the interval within which event timing is observed, while within-exposure event timing contains illumination-locked structure and therefore cannot be uniquely attributed to object timing without separating that

structure. DSEC [15] publishes per-frame exposure meta-data, and the files resolve for eighteen train sequences. The width varies by a factor of 127 against a frame period fixed at 50 ms, and within one recording by up to a factor of 12.5. The auto-exposure ceiling of  $14\,996\,\mu\text{s}$  holds 32.5 % of frames, in six of the eighteen sequences, and the distribution is not continuous over that range (Fig. 3). The gap separates the two exposure regimes, and ceiling exposure forms a distinct stratum.

The released DSEC-Det labels comprise 390 118 boxes on 6693 tracks across 60 sequences. Every label sits on the frame clock, and the median spacing between consecutive label times is  $50\,001\,\mu\text{s}$ . Over 361 628 triples of consecutive labels on a track, 7.1 % of middle box centers fall within a hundredth of a pixel of their neighbors’ linear interpolant. This rules out exact local linear interpolation as a general description of the released trajectories without identifying the underlying annotation process. DSEC-Det ships more than one annotation version [18], the supplement names the one measured here. Median box-center image speed is 40 px/s, with 360 px/s at the ninety-ninth percentile.

### 5.2. Mains-frequency signature and the per-pixel phase test

Swiss mains is 50 Hz, and rectified lighting can produce the tested 100 Hz modulation. The pre-specified 100-Hz line and its harmonics appear in the event-rate spectrum of every ceiling sequence, while the daytime control lacks the same pattern (Fig. 4). Per-pixel phase tests and the pre-specified frequency control establish the same illumination signature.

Per pixel, the Rayleigh statistic  $Z = R^2/N$  is  $\text{Exp}(1)$  under independent uniform phases. The off-frequency and daytime controls empirically calibrate this nominal null, while the ceiling sequence has median  $Z = 2.935$  at 100 Hz and 24.07 % of live pixels reach the nominal  $p < 10^{-3}$  threshold under the independent-uniform-phase reference. The illumination attribution therefore relies primarily on the frequency-matched and daytime controls, not on the nominal Rayleigh reference alone.

### 5.3. All six ceiling sequences, per labeled box

The same statistic taken per labeled box, over the events inside that box during that frame’s own exposure window, extends to every ceiling sequence. Across the six sequences and 16 516 boxes, 95.4–100.0% satisfy the pre-specified criterion in Supplement Table 2.

### 5.4. Within-box temporal dispersion under that signature

That signature limits attribution of within-exposure event timing to object timing. The per-box 137-Hz arm is used as a same-event descriptive control, not as an independent spectral-resolution test. The illumination attribu-

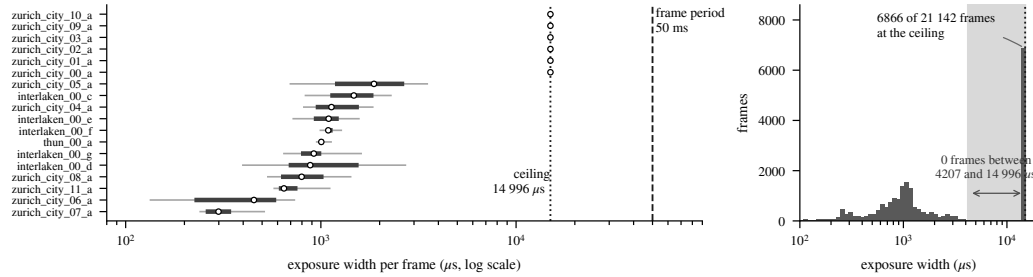


Figure 3. Published per-frame exposure width across the eighteen DSEC training sequences whose exposure files resolve: per sequence (left, log axis, median, interquartile range and 5–95 % band, against the fixed 50 ms frame period) and pooled over 21 142 frames (right). 32.5 % of frames sit at the auto-exposure ceiling of 14 996  $\mu$ s and zero fall between the largest non-ceiling width of 4207  $\mu$ s and that ceiling, separating the two exposure regimes. The six ceiling sequences are those measured in Sec. 5.

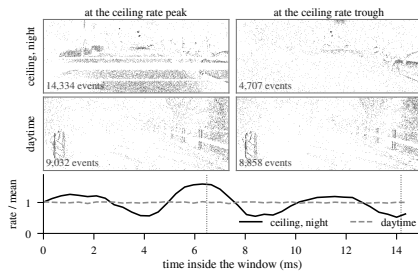


Figure 4. Top: 350  $\mu$ s slices of the ceiling-exposure sequence `zurich_city_09_a` and of the daytime control, at the peak and trough of the ceiling sequence's own event rate inside its densest published exposure window. Bottom: that rate in 350  $\mu$ s bins, each sequence normalized to its own mean. The ceiling rate moves by a factor of 3.0 between its extremes, the daytime rate by 1.1.

required for that counterexample.

## 7. Conclusion

In the released SSM-ViT pipeline one pooled score combines predictions with one to twenty-one windows of recurrent history, and moving only the chunk boundary shifts S5-B by 4.20 mAP over all labeled validation boxes and 5.23 mAP on the velocity-evaluable subset, at fixed frames, weights, labels and evaluator. The chunk boundary is an unreported degree of freedom of the protocol: it changes the recurrent support behind the reported score while every recorded quantity stays fixed, which is exactly a failure of temporal-support identifiability.

The result does not single out a canonical temporal-support statistic, nor attribute the shift to one stage of the pipeline. RVT's input sensitivity and its regression-based temporal coefficient remain descriptors rather than estimates of recurrent support, and DSEC's within-exposure timing carries illumination-locked structure that must be separated before it is attributed to object motion.

Two scores at the same nominal timestamp are not performance at the same effective time when the recurrent support behind them differs, so comparisons across recurrent architectures should report or control predictor initialization, reset policy, chunk boundaries, effective history and label-acquisition intervals alongside the score, making them protocol-identifiable.

tion is supported primarily by the long-slice 100-Hz harmonic structure and the daytime control. Within-box dispersion is 206.6  $\mu$ s against a null of 93.0  $\mu$ s, with an excess of 183.6  $\mu$ s. A frame-level permutation preserving pooled temporal distributions and box counts gives a null of 78.6  $\mu$ s, and the excess remains after the clean-pixel control. Random-position boxes show that the dispersion is not specific to annotated object regions (Supplement Sec. 9), which supports reading the signature as an attribution limit and not as object motion.

## 6. Scope and limitations

Each supporting audit is conditional on its stated instrument: local window-ablation sensitivity, nominal-time association under the reported regression specification, and a local constant-velocity displacement surrogate. The primary SSM chunk-boundary intervention is established on two released SSM-ViT checkpoints under their released streaming evaluator; the second tests that the violation is not confined to one weight set. Gen1 and DSEC are separate audits of predictor- and label-side timing, not premises

## References

- [1] W.-C. Hung, V. Casser, H. Kretzschmar, J.-J. Hwang, and D. Anguelov. LET-3D-AP: Longitudinal error tolerant 3D average precision for camera-only 3D detection. *arXiv:2206.07705*, 2022. Waymo Open Dataset 3D camera-only detection challenge metric. 3
- [2] S. Tanaka, K. Minoda, F. Watanabe, and T. Horibe. Rethink 3D object detection from physical world. *arXiv:2507.00190*, 2025.
- [3] A. Kugele, T. Pfeil, M. Pfeiffer, and E. Chicca. How many events make an object? Improving single-frame object detection on the 1 Mpx dataset. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2023, pages 3913–3922.
- [4] Z. Zhou, Z. Wu, R. Bouteau, F. Yang, C. Demonceaux, and D. Gin-hac. RGB-Event fusion for moving object detection in autonomous driving. *arXiv:2209.08323*, 2022.
- [5] Waymo Open Dataset. `label.proto`, `field camera_synced_box` and its documentation comment, read 2026. 2
- [6] P. Sun et al. Scalability in perception for autonomous driving: Waymo Open Dataset. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [7] A. J. Miro, L. af Klinteberg, B. Timus, A. Asefaw, A. Khoche, T. Gustafsson, S. S. Mansouri, and M. Daneshmand. Correcting and quantifying systematic errors in 3D box annotations for autonomous driving. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. 2
- [8] T. Qin and S. Shen. Online temporal calibration for monocular visual-inertial systems. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018. 5
- [9] M. Li, Y.-X. Wang, and D. Ramanan. Towards streaming perception. In *European Conference on Computer Vision (ECCV)*, 2020. 2
- [10] H. Mao, X. Yang, and W. J. Dally. A delay metric for video object detection: What average precision fails to tell. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pages 573–582.
- [11] D. Bolya, S. Foley, J. Hays, and J. Hoffman. TIDE: A general toolbox for identifying object detection errors. In *European Conference on Computer Vision (ECCV)*, 2020. 3
- [12] D. Hoiem, Y. Chodpathumwan, and Q. Dai. Diagnosing error in object detectors. In *European Conference on Computer Vision (ECCV)*, 2012.
- [13] H. Alwassel, F. Caba Heilbron, V. Escorcia, and B. Ghanem. Diagnosing error in temporal action detectors. In *European Conference on Computer Vision (ECCV)*, 2018.
- [14] J. Luiten, A. Oşep, P. Dendorfer, P. Torr, A. Geiger, L. Leal-Taixé, and B. Leibe. HOTA: A higher order metric for evaluating multi-object tracking. *International Journal of Computer Vision*, 2021. 3
- [15] M. Gehrig, W. Aarents, D. Gehrig, and D. Scaramuzza. DSEC: A stereo event camera dataset for driving scenarios. *IEEE Robotics and Automation Letters*, 2021. 7
- [16] D. Gehrig and D. Scaramuzza. Low-latency automotive vision with event cameras. *Nature*, 629:1034–1040, 2024. 3
- [17] J. Pang, L. Qiu, X. Li, H. Chen, Q. Li, T. Darrell, and F. Yu. Quasi-dense similarity learning for multiple object tracking. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- [18] K. Wang, S. Liu, H. Shi, L. Shi, and H. Chen. Beyond duality: A hybrid framework of leveraging shared and private features for RGB-Event object detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. 7
- [19] P. de Tournemire, D. Nitti, E. Perot, D. Migliore, and A. Sironi. A large scale event-based detection dataset for automotive. *arXiv:2001.08499*, 2020. 3
- [20] E. Perot, P. de Tournemire, D. Nitti, J. Masci, and A. Sironi. Learning to detect objects with a 1 megapixel event camera. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [21] H. Cho, J.-Y. Kang, Y. Kim, and K.-J. Yoon. Ev-3DOD: Pushing the temporal boundaries of 3D object detection with event cameras. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 3
- [22] L. Sun, C. Sakaridis, J. Liang, Q. Jiang, K. Yang, P. Sun, Y. Ye, K. Wang, and L. Van Gool. Event-based fusion for motion deblurring with cross-modal attention. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2
- [23] T. Kim, J. Lee, L. Wang, and K.-J. Yoon. Event-guided deblurring of unknown exposure time videos. In *European Conference on Computer Vision (ECCV)*, 2022. 2
- [24] M. Gehrig and D. Scaramuzza. Recurrent vision transformers for object detection with event cameras. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2, 4
- [25] H. Liu, S. Peng, L. Zhu, Y. Chang, H. Zhou, and L. Yan. Seeing Motion at Nighttime with an Event Camera. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2
- [26] N. Zubić, M. Gehrig, and D. Scaramuzza. State space models for event cameras. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2, 5
- [27] Z. Wu, M. Gehrig, Q. Lyu, X. Liu, and I. Gilitschenski. LEOD: Label-efficient object detection for event cameras. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16933–16943, 2024. 3
- [28] Z. Sui, H. Hao, W. He, S.-H. Lee, and W. Wang. Adaptive spatial-temporal window: unlocking the potential of event cameras in heterogeneous velocity scenarios. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 946–955, 2026. 2
- [29] Y. Yang, J. Liang, B. Yu, Y. Chen, J. S. Ren, and B. Shi. Latency correction for event-guided deblurring and frame interpolation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24977–24986, 2024. 2
- [30] J. Chu, R. Zhang, C. Yang, Z. Yu, Z. Bu, H. Liu, F. Röhrbein, A. Knoll, G. Chen, and C. Jiang. Bridging the latency gap with a continuous stream evaluation framework in event-driven perception. *Nature Communications*, 17:2441, 2026. 2
- [31] S. Kuzucu, K. Oksuz, J. Sadeghi, and P. K. Dokania. On calibration of object detectors: Pitfalls, evaluation and baselines. In *European Conference on Computer Vision (ECCV)*, 2024. 2
- [32] R. Fogliato, P. Patil, M. Monfort, and P. Perona. A framework for efficient model evaluation through stratification, sampling, and estimation. In *European Conference on Computer Vision (ECCV)*, 2024. 3
- [33] C. G. Northcutt, A. Athalye, and J. Mueller. Pervasive label errors in test sets destabilize machine learning benchmarks. In *NeurIPS Datasets and Benchmarks*, 2021.
- [34] B. Ercan, O. Eker, A. Erdem, and E. Erdem. EVREAL: Towards a comprehensive benchmark and analysis suite for event-based video reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2023. 2
- [35] J. Yang, S. Liu, Z. Li, X. Li, and J. Sun. Real-time object detection for streaming perception. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. 2
- [36] R. Salles et al. SoftED: Metrics for soft evaluation of time series event detection. *Computers & Industrial Engineering*, 2024. 2
- [37] Haitian Zhang et al. Frequency-adaptive low-latency object detection using events and frames. *arXiv:2412.04149*, 2024. Unrefereed preprint. 2